

HOW MANY SHAFTS DOES IT TAKE TO WEAVE A FACE

Draft N040: 40 shafts, 38 pattern blocks. Taquete (weft-faced compound tabby).

Julia Jackson, photographed by Julia Margaret Cameron in 1867 (The Met, public domain), reduced to two wefts on 40 shafts.

Structure. Warp ends alternate pattern, binding, pattern, binding. Binding ends go on shafts 1 and 2 alternately. Each pattern end belongs to one block; block b is shaft $b+2$. Every pick lifts one binding shaft (1, 2, 1, 2 ...) plus the pattern blocks that must hide THIS weft. Two wefts per pass, white and black, and the order alternates each pass (W K, K W) so the binding points checker.

Counts. 480 ends (240 pattern + 240 binding), 1200 picks = 600 passes = 300 pattern rows woven two passes each. Image grid 240 x 300.

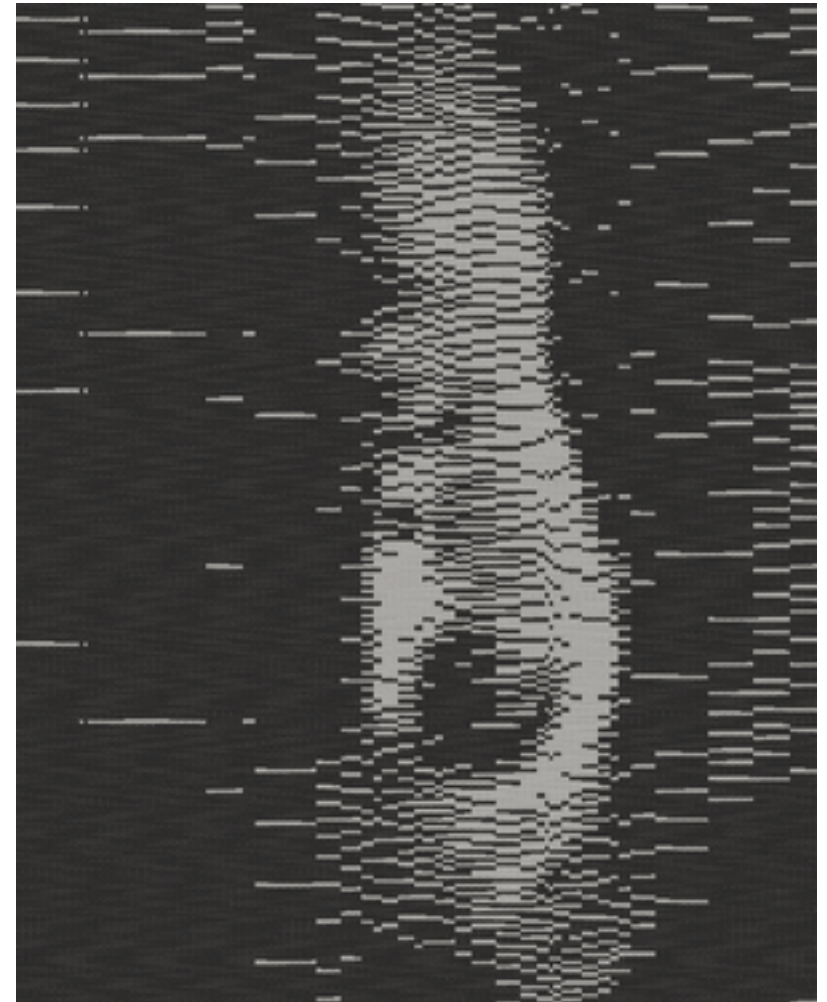
Sett. Weft-faced: the warp should not show. Something like 10/2 cotton or linen warp at 20 to 24 epi (8 to 10 per cm), wefts noticeably thicker than the warp (a 3/2 or 5/2 cotton, or two strands), beaten hard so each pass covers about as much length as two ends do width. Check the first 40 passes: a 20-row test square should come out square. If it is short, beat lighter or use a fatter weft.

Floats. By construction no weft floats over more than three ends and no warp end floats over more than two picks, so the cloth is sound at any block count.

The back is the negative. The weft that hides on the face shows on the back.

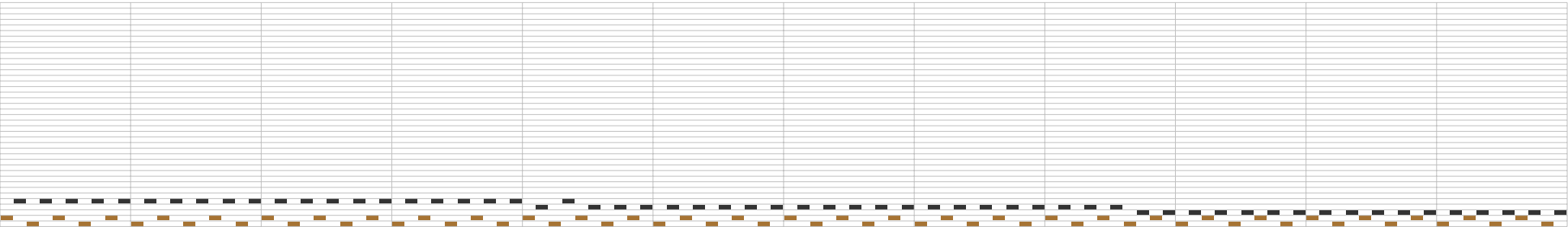
Files: julia_N040.wif (this draft, liftplan form, loads in Fiberworks, WeavePoint, PixeLoom, etc.).

Threading strips follow; the liftplan is drawn on the last pages as a reference, read it from the WIF for weaving.

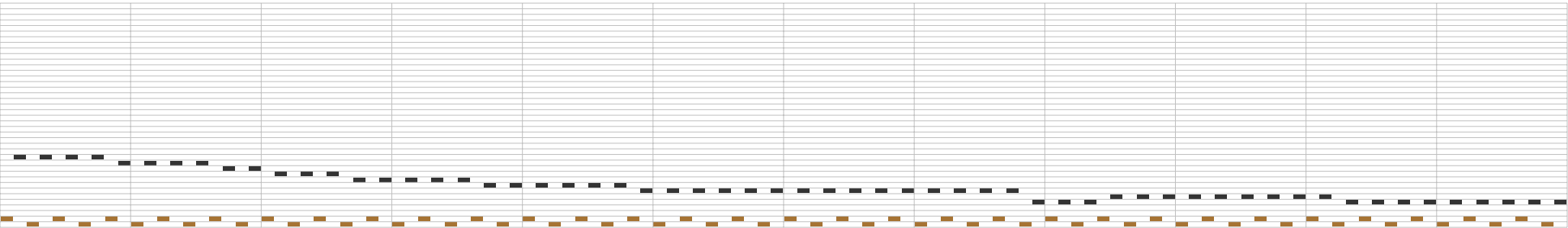


Threading, 40 shafts. Read right to left as weavers do: end 1 is at the right of the first strip. Shaft 1 is the bottom row.

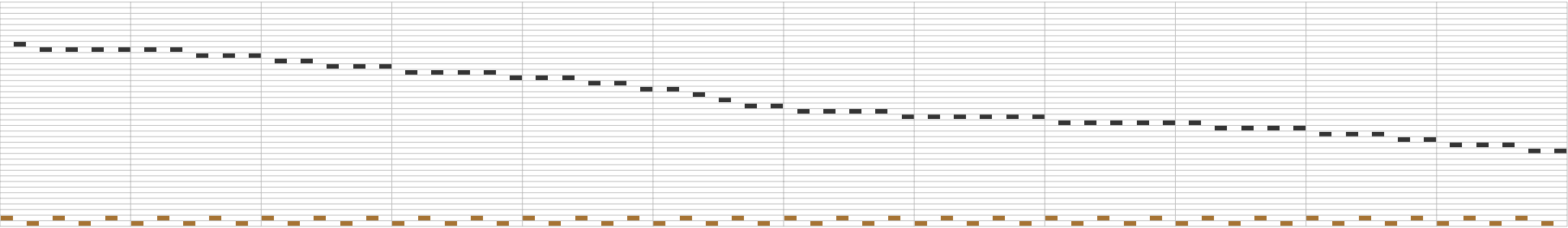
ends 1 to 120



ends 121 to 240



ends 241 to 360

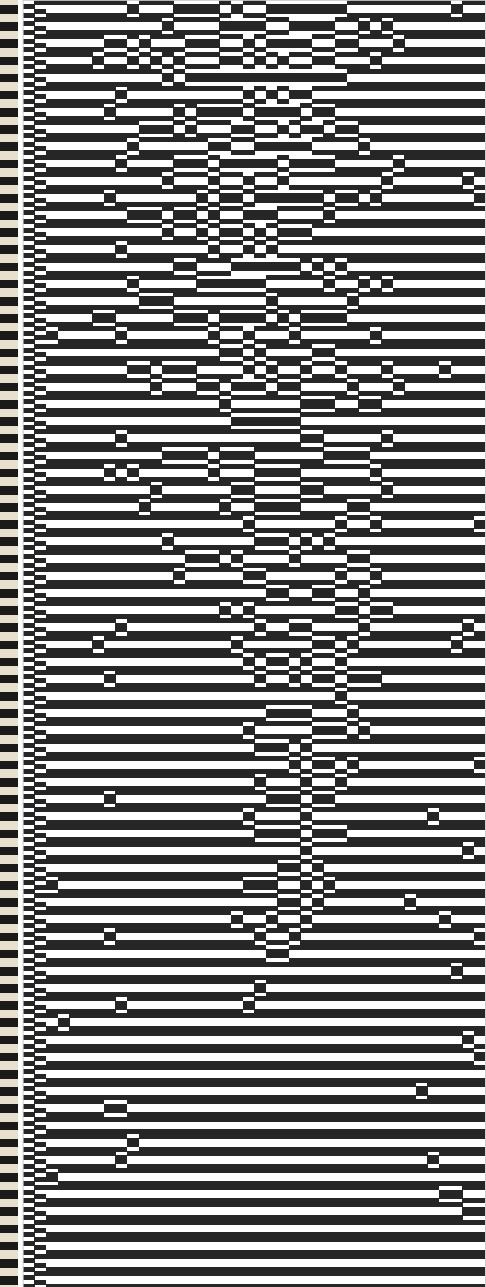


Threading, 40 shafts. Read right to left as weavers do: end 1 is at the right of the first strip. Shaft 1 is the bottom row.

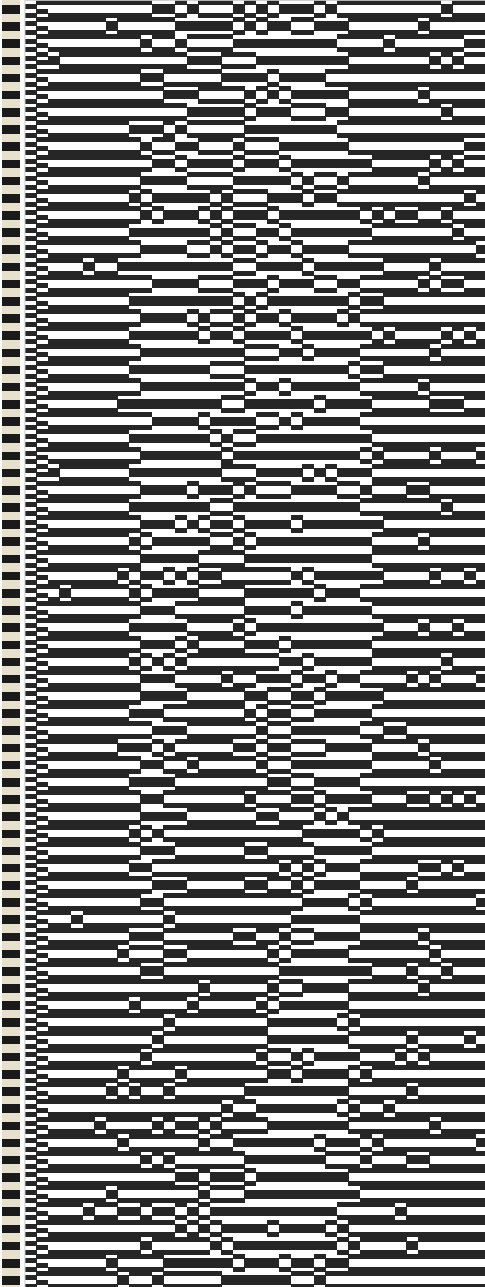
ends 361 to 480

Year	Men (%)	Women (%)
1975	95	95
1980	96	96
1985	97	97
1990	96	97
1995	96	97
2000	95	97
2005	95	97
2010	94	97
2015	90	95

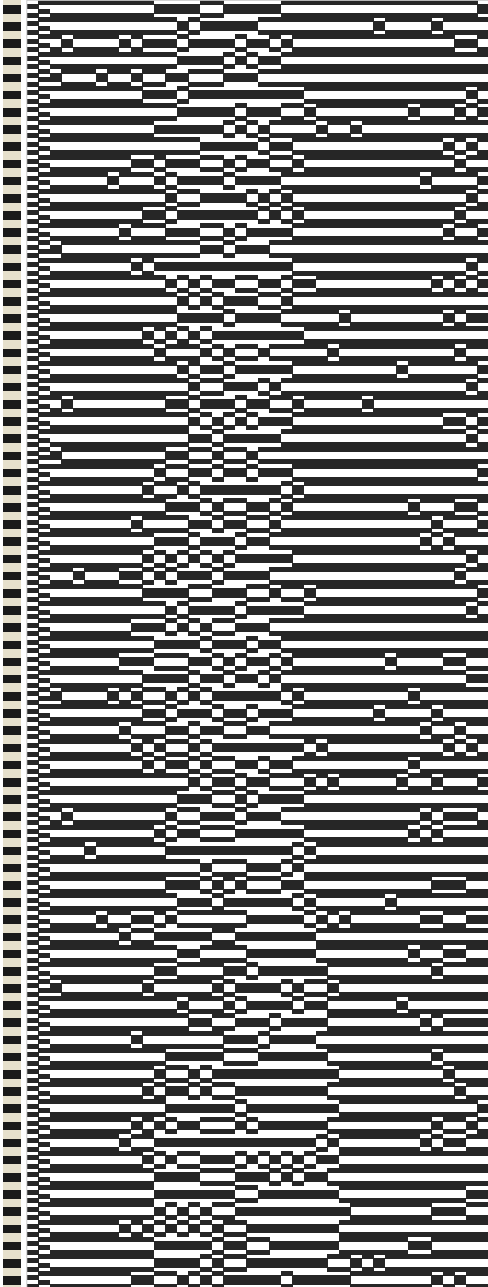
Liftplan, 40 shafts, 1200 picks, bottom to top in each column, columns left to right. Filled = shaft lifted. Left margin: weft colour.



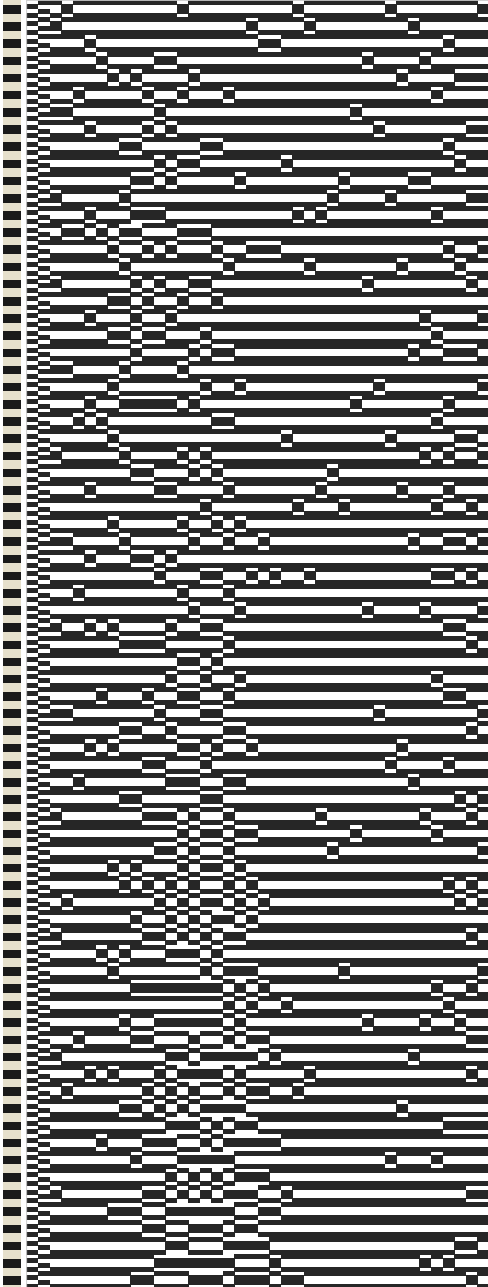
picks 1 to 300



picks 301 to 600



picks 601 to 900



picks 901 to 1200